



DIVE INTO AI

LLM-LIKE INTERACTIONS THROUGH CHATBOTS

Güldeniz Bektaş,

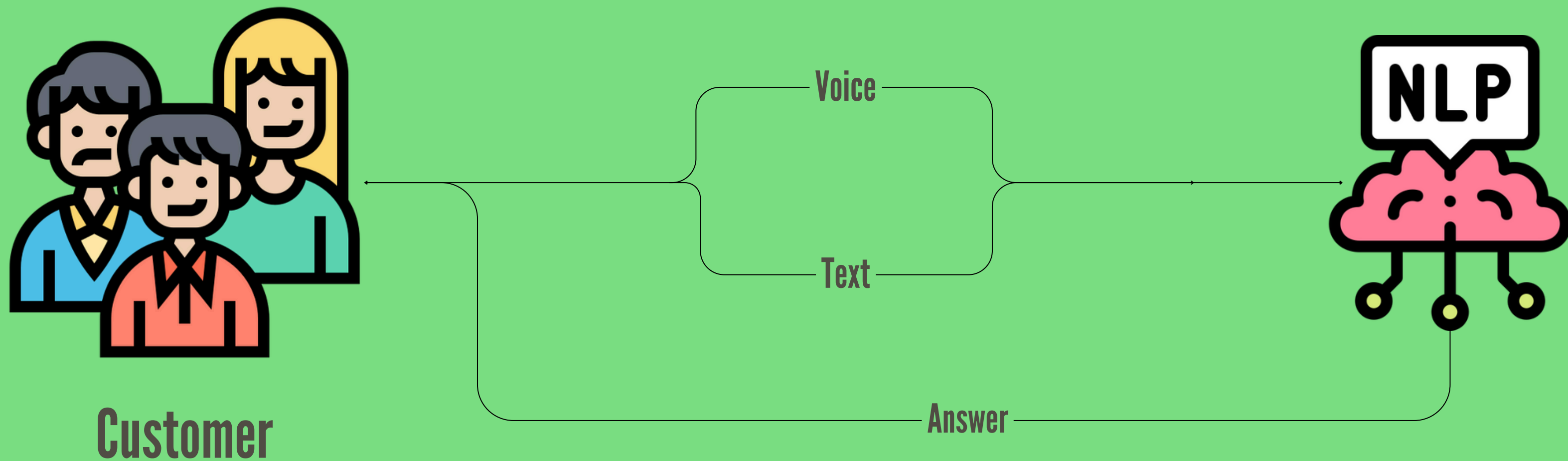


KNOW ME

AI Engineer
Astronom
#MoreThan101

TurkNet
Istanbul University
Founder

TASK - CHATBOTS



PATTERN MATCHERS

“AIML” (artificial intelligence markup language)

```
<aiml version = "1.0.1" encoding = "UTF-8"?>
```

```
<category>
```

```
<pattern> WHO IS ALBERT EINSTEIN </pattern>
```

```
<template>Albert Einstein was a German physicist.</template>
```

```
</category>
```

```
<category>
```

```
<pattern> WHO IS Isaac NEWTON </pattern>
```

```
<template>Isaac Newton was a English physicist and mathematician.</template>
```

```
</category>
```

```
<category>
```

```
<pattern>DO YOU KNOW WHO * IS</pattern>
```

```
<template>
```

```
<srai>WHO IS <star/></srai>
```

```
</template>
```

```
</category>
```

```
</aiml>
```

Simple output

Do you know who Albert Einstein is?



Albert Einstein was a German physicist.

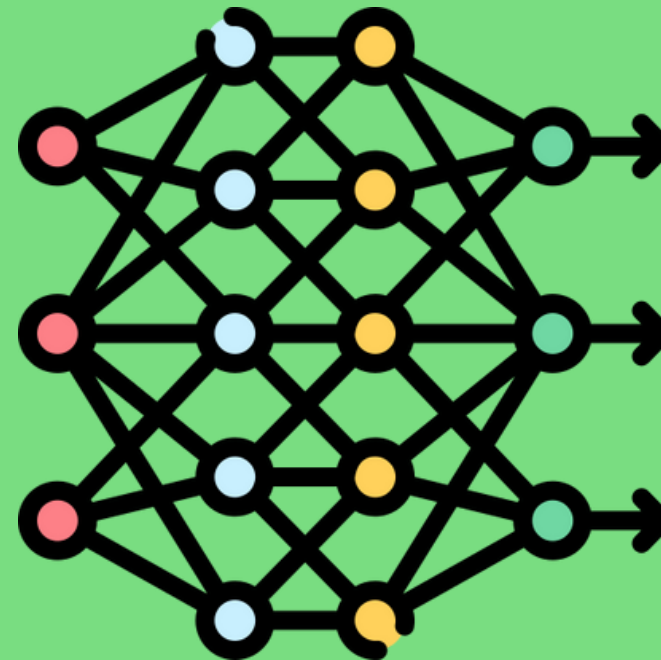


LARGE LANGUAGE MODELS

train on huge sets of data

LLMs are built on neural network called a transformer model

interpreting questions and generating responses, or translating text from one language to another



REINFORCEMENT LEARNING FROM HUMAN FEEDBACK (RLHF)

Reinforcement learning from human feedback (RLHF) is a machine learning (ML) technique that uses human feedback to optimize ML models to self-learn more efficiently.

But LLMs have limitations and may not fully understand the user request. The question may be too open-ended, or the person may not be clear enough in their instructions. To teach ChatGPT how to create dialogue in a human style of conversation, it was trained using RLHF so the AI learns human expectations.

1. Data Collection

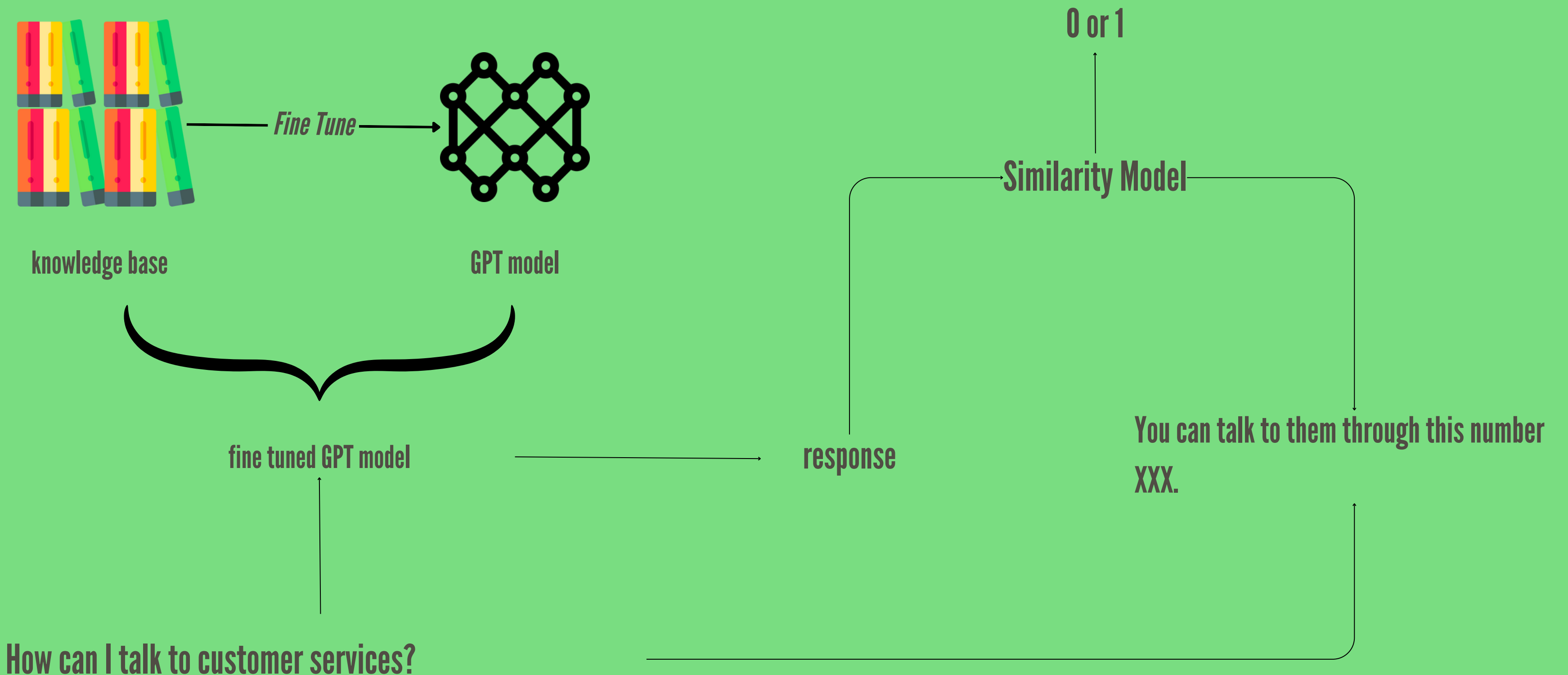
"How can I talk to customer services?"

"How can I pay my invoice?"

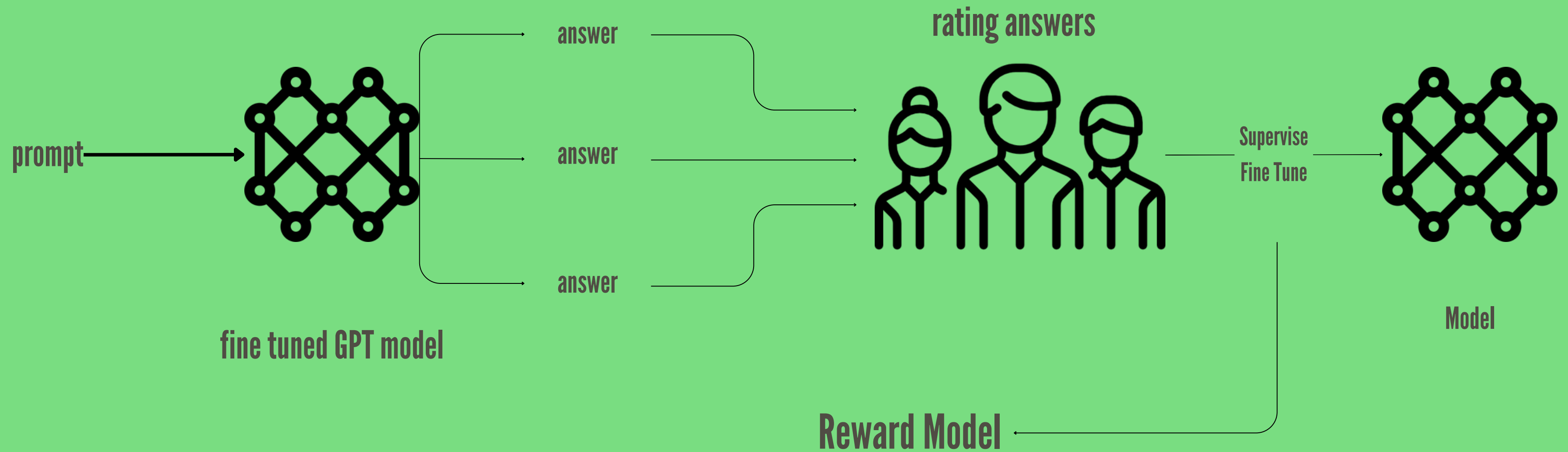
"LET ME TALK TO A REAL PERSON!"



2. Supervised Fine-Tuning of the Language Model

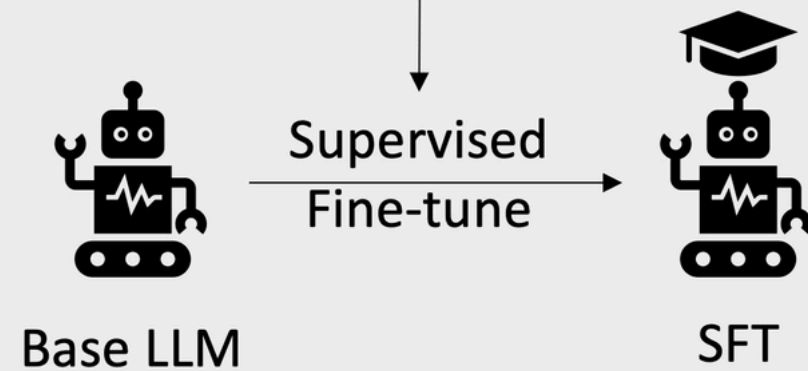


3. Building a Separate Reward Model:

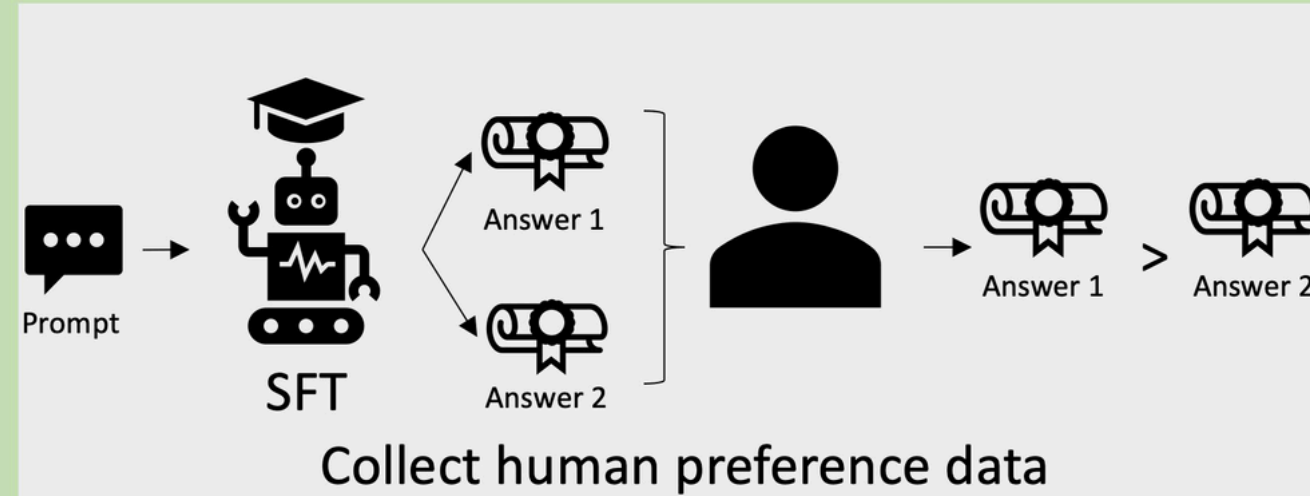


4. Optimizing the Language Model with the Reward-Based Model:

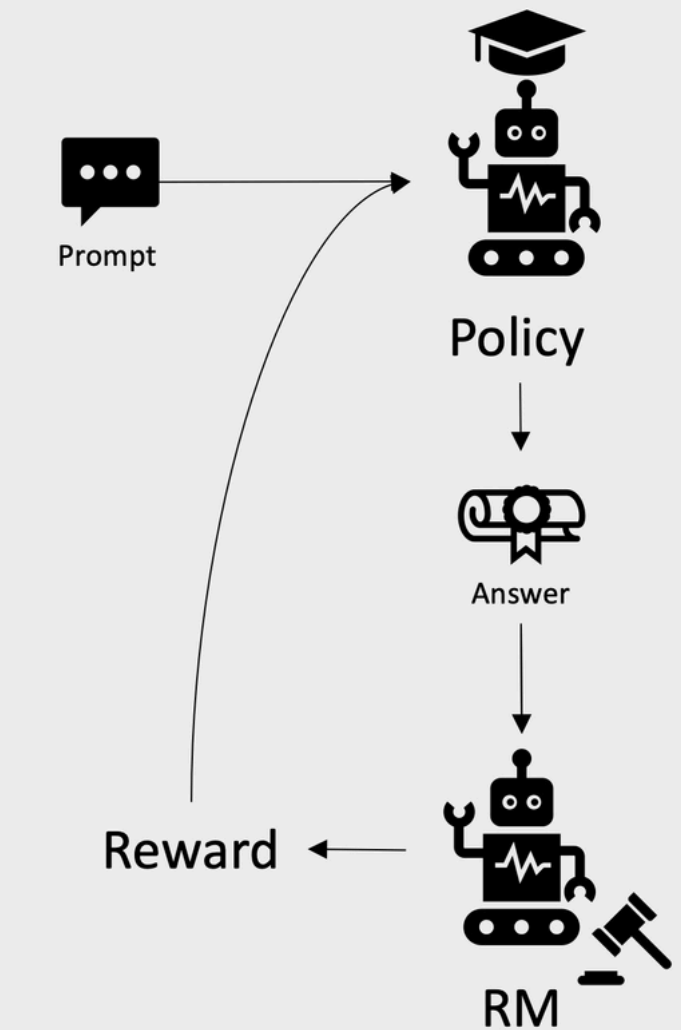
Step 1 Supervised Fine-Tuning



Step 2 Training a Reward Model



Step 3 Optimize Policy



PPO

RAG - RETRIEVAL AUGMENTED GENERATION

- Limited knowledge
- Private data
- Specific information

A typical RAG application has two main components:

1. Indexing: a pipeline for ingesting data from a source and indexing it. This usually happens offline.
2. Retrieval and generation: the actual RAG chain, which takes the user query at run time and retrieves the relevant data from the index, then passes that to the model.

PROCESS OF KNOWLEDGE BASE

Load



Split



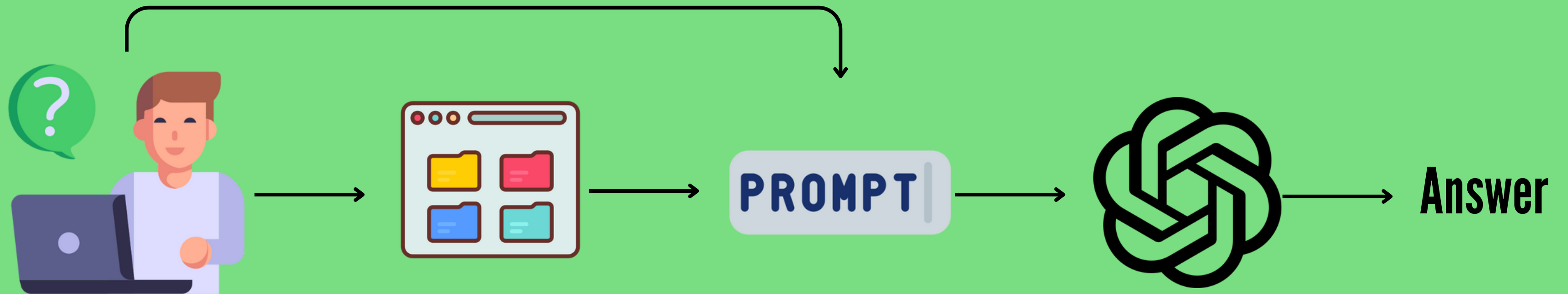
Embed



Store



PROCESS OF RAG APPLICATION



MORE THAN ONE KB?

AGENT-TOOLS



Which tool to use? /
Which knowledge to use?





Thanks!
You're cool.



SCAN ME

Contact Me